FACMIC

FEDERATED ADAPTATIVE CLIP MODEL
FOR
MEDICAL IMAGE CLASSIFICATION

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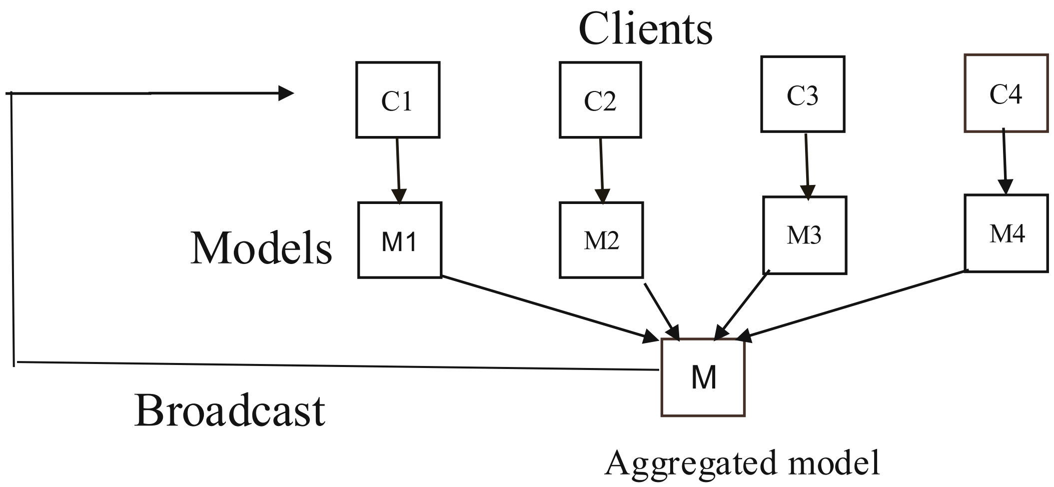
SUMMARY (Previous presentation)

Federated Learning
- Data Privacy & Security
- Decentralized data access
- Aggregation on model level : no data sharing



Influencing factors in FL
- Communication cost
- Data distribution shifts across clients

VLMs like Clip have too many parameters – Huge communication cost

Requirements for lighter models which can adapt to different domains or clients

We want models which can adapt to real life data as well

SUMMARY (Previous presentation)

PROBLEM STATEMENT

FACMIC Setting:

- We consider N clients (e.g. Hospitals) namely $C_1, C_2 \ldots, C_N$
- Each client holds a private medical image dataset – i-th client C_i has dataset $D_i = \{(x_{i,j}, y_{i,j})\}$, $j=1,2,\ldots,n_i$
- Data cannot be shared among clients.
- We assume that all clients have same input and output space but the data distribution are different for different clients i.e. $P(D_i) \neq P(D_{i'})\ for\ all\ i \neq i'$
- Each dataset D_i consists of three non-overlapping subsets- Training set (D_i^{train}), Validation set (D_i^{val}) & Test set (D_i^{test})
- We want to train a robust global model $f_\theta(.)$ that preserves data privacy of each client
- This model should perform well on test data of each of the clients i.e.

$$\min_f \frac{1}{N} \sum_{i=1}^{N} \frac{1}{n_i^{test}} \sum_{j=1}^{n_i^{test}} l\left(f_\theta\left(x_{i,j}^{test}\right), y_{i,j}^{test}\right)$$

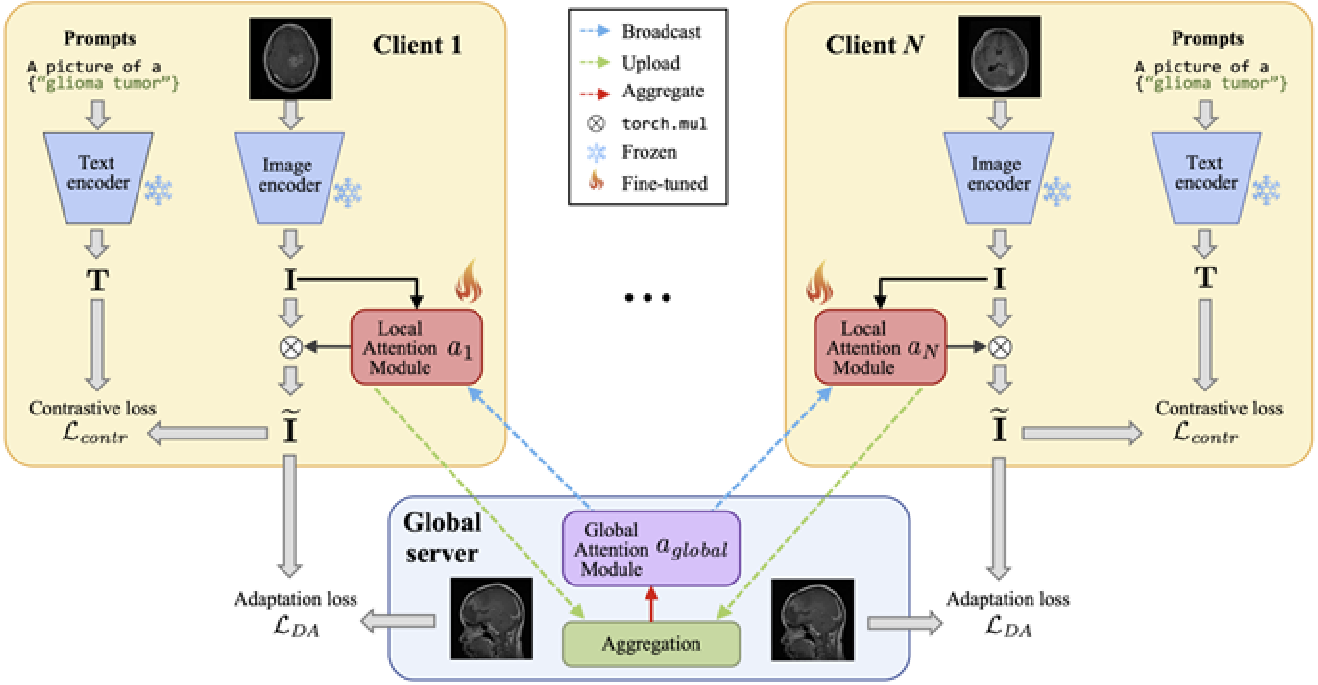
- Also, this model should perform well on any client’s data which were excluded from training stage.

SUMMARY (Previous presentation)

METHODOLOGY

FACMIC Framework

Federated Adaptative CLIP Model for Medical Image Classification



- Uses a pre-trained CLIP model with an image encoder & text encoder
- FACMIC adds a light-weight feature attention module on top of CLIP model
- Also introduces a domain adaptation (DA) strategy to minimize data discrepancies between clients
- Image and text encoders are frozen- only the attention module is trained

SUMMARY (Previous presentation)

EXPERIMENTS

DATASET

BT (Brain Tumor)

- 4 classes
- 2870 training samples
- 394 testing samples
- 100 training rounds

SC (Skin cancer)

- Obtained from ISIC
- 9 classes
- 2239 training samples
- 118 testing samples
- 50 training rounds

Real multi-source skin cancer

- ISIC2019+ SC+HAM10000
- 7 classes
- 30000 training samples
- 6233 testing samples
- 50 training rounds

CLIENT DIVISION

- **IID distribution**-for BT,SC Divide training samples into 3 equal sizes as 3 clients
- **Non-IID distribution** – for BT,SC use Dirichlet distribution, with concentration parameter $\alpha = \{0.3, 0.6, 0.9\}$ to generate non-iid distribution among clients-

Samples of each client under non-iid conditions for BT and SC datasets.

Clients	$\alpha = 0.3$		$\alpha = 0.6$		$\alpha = 0.9$	
	BT	SC	BT	SC	BT	SC
Client1	2075	218	1355	387	1480	1156
Client2	389	1374	908	487	423	335
Client3	406	647	607	1365	967	748
Global	394	118	394	118	394	118

- **For the Real dataset-** Client 1 (SC) has 1971 samples, Client 2 (ISIC2019) has 19766 samples, Client 3 (HAM10000) has 8512 samples. Different sources act as different clients

SUMMARY (Previous presentation)
RESULTS

❖ Authors compare the results of FACMIC with state-of-the-art methods – FedAVG, MOON, FedProx, FedFocal, FedCLIP & a centralized method.

❖ Accuracy metric (ACC) is used for all the datasets. In the real dataset due to class imbalance Balanced Accuracy (average recall across classes) has also been used along with accuracy.

❖ Results for different methods are shared below-

Method	$\alpha = 0.3$		$\alpha = 0.6$		$\alpha = 0.9$		*iid*		Average		Real	
	BT	SC	BT	SC	BT	SC	BT	SC	BT	SC	ACC	BACC
Centralized	✘	✘	✘	✘	✘	✘	✘	✘	70.30	**61.01**	60.81	47.80
FedAVG	66.24	48.30	66.49	49.15	64.21	54.23	68.52	49.15	66.36	50.21	60.24	50.96
FedProx	66.24	48.30	63.95	51.69	65.23	49.15	70.05	50.84	66.37	49.99	59.52	48.89
MOON	67.25	50.84	63.96	50.00	65.48	53.38	69.29	47.45	66.49	50.42	59.95	50.90
FedFocal	67.00	49.15	65.48	49.15	67.26	54.23	64.21	51.69	65.99	51.05	59.97	49.80
FedCLIP	68.78	53.38	66.24	55.93	67.26	54.23	66.50	50.0	67.19	53.38	58.43	51.17
Ours	**82.74**	**56.78**	**82.23**	**58.47**	**81.73**	**56.78**	**82.23**	**57.62**	**82.23**	57.41	**72.37**	**63.61**

❖ FACMIC outperforms all the existing methods by quite good margin. The following graph shows the accuracy on the BT data-

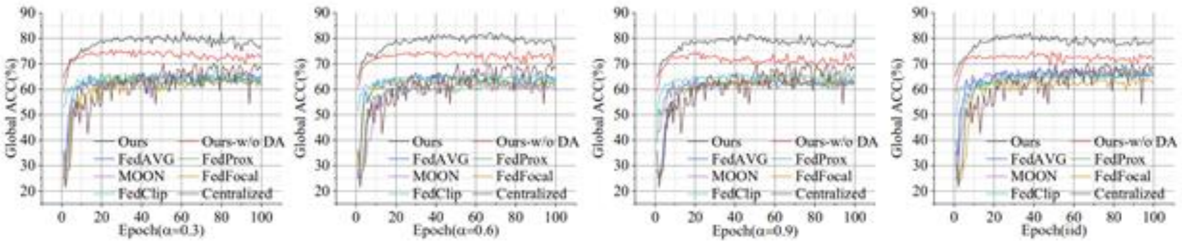
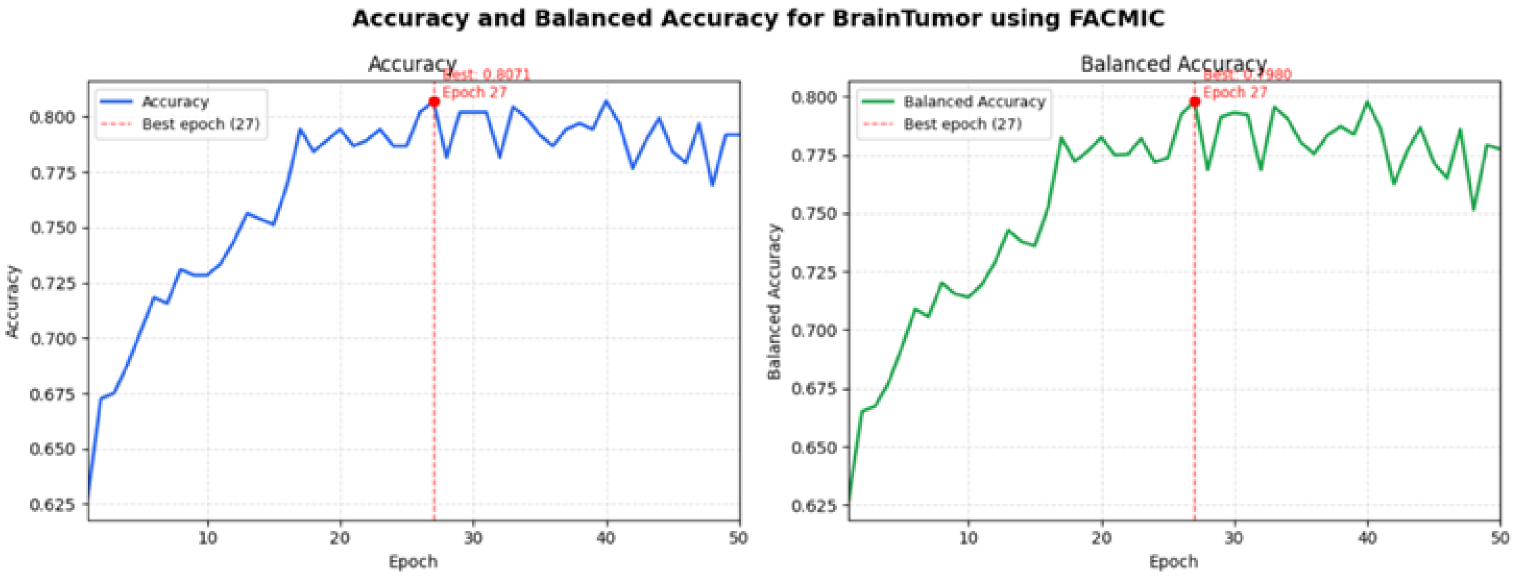


Fig. 2: Global testing accuracy (%) for each round on the BT dataset.

RESULTS FROM IMPLEMENTATION

The data provided is the Brain Tumor dataset in the Non-IID setting ,with alpha = 0.9. The code is run in the same setup without any change

```
=== Accuracy ===
  First  (Epoch  1): 0.626904
  Best   (Epoch 27): 0.807107
  Final  (Epoch 50): 0.791878

=== Balanced Accuracy ===
  First  (Epoch  1): 0.627265
  Best   (Epoch 27): 0.798041
  Final  (Epoch 50): 0.777444
```



Best accuracy : 80.71 % , Best balanced accuracy: 79.80%

NOVEL IDEAS

Dynamic prompts - Instead of fixed prompts :

Fixed Prompts

- FACMIC uses CLIP model which uses static prompts . Eg.- ***"An image showing glioma"***
- prompts are manually designed and remain the same for all images within a class, regardless of variations in tumor shape, size, or location.
- *Intra-class variability* is not considered
- This may lead to –
    - Weak image-text alignment
    - Loss of semantic information
    - Reduced performance in contrastive learning
- the model learns a coarse mapping rather than a clinically meaningful one.

Dynamic Prompts

- Instead, we generate dynamic prompts i.e. prompts describing each image. So, prompts become something like **- *"brain MRI showing infiltrative glioma with irregular borders and edema".***
- Prompt reflects visual characteristics & image-specific radiological features.
- Each image gets a unique textual description which leads to better matching in embedding space
- Instead of a single label, the model sees: structure, context & clinical attributes
- Model learns features, not just labels, which improves generalization

NOVEL IDEAS

Dynamic Prompt Generation:

Dataset structure

- Brain tumor data
  - client_0
    - glioma_tumor
      - img_1.jpg
      - img_2.jpg
    - meningioma_tumor
    - no_tumor
    - pituitary_tumor
  - client_1 (same structure)
  - client_2 (same structure)
  - client_3 (same structure)

Llama 3.2 pipeline

Raw image ingestion

All image paths are stored in a list and fed into the pre-trained llama 3.2 model

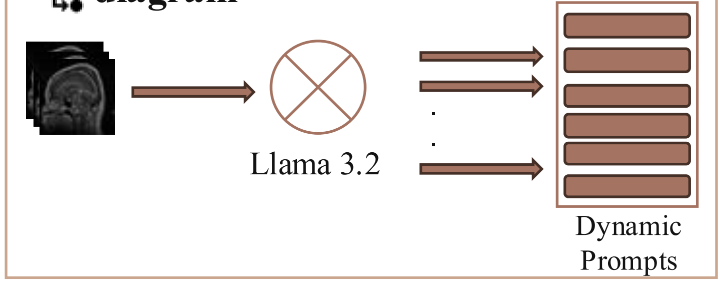
Vision-Language Processing

Llama 3.2 is instructed to behave like a radiologist. It analyzes pixel data to identify tumor characteristics (location, size, texture).

Prompt Generation

Image-specific prompts are generated. Such as – *“Brain MRI showing diffuse, perivascular glioma with scattered cortical infiltrates.”*

diagram



Why Llama 3.2

Domain specific language generation

LLaMA 3.2 provides strong language understanding and generation capability, allowing it to produce coherent and medically relevant descriptions.

Efficiency compared to other models

it is computationally efficient compared to larger proprietary models, making it practical for large-scale prompt generation across thousands of medical images..

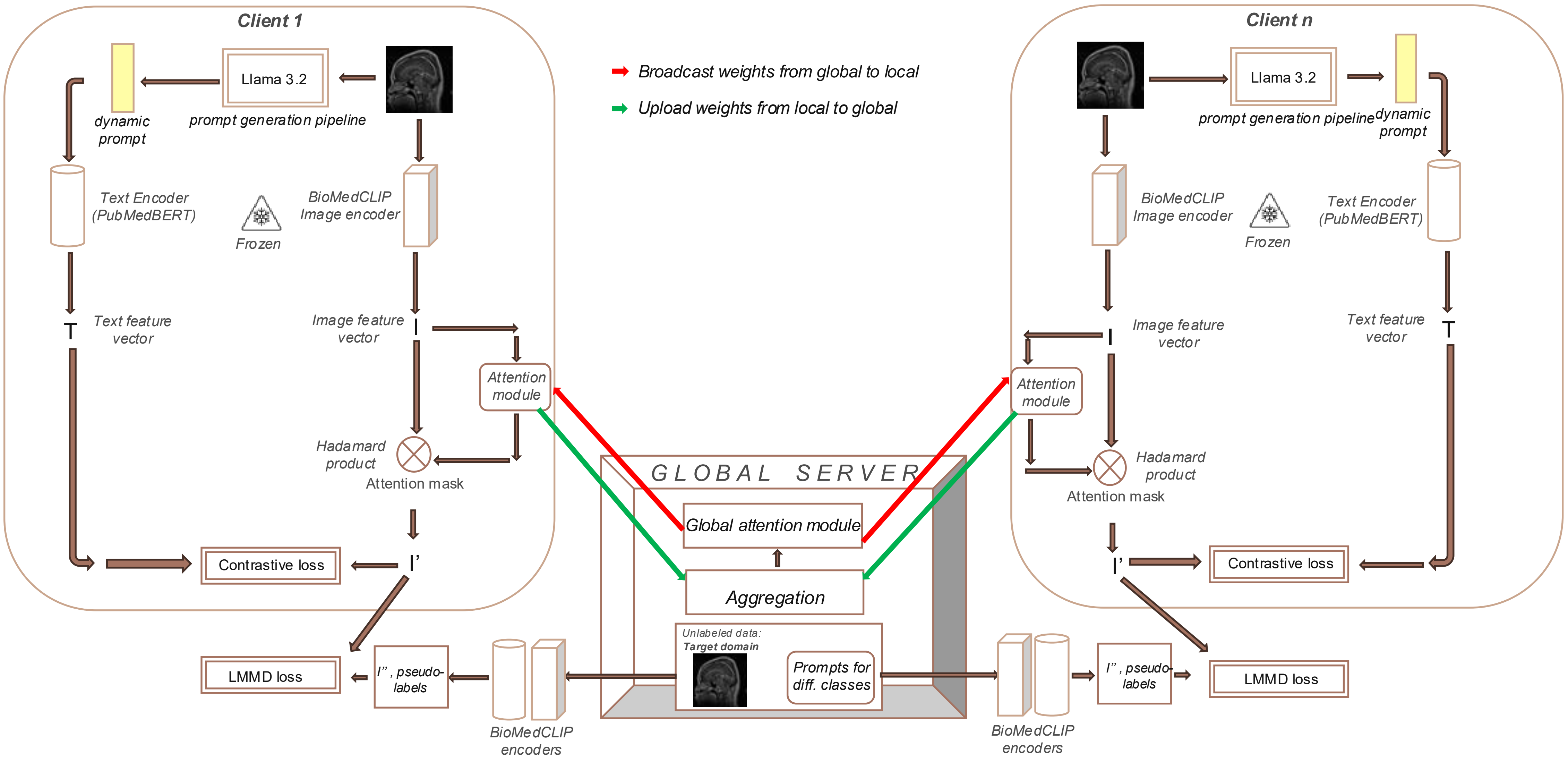
NOVEL IDEAS

Addressing limitations of CLIP using BioMedCLIP

CLIP	BioMedCLIP
Training Data: • WebImage Text (WIT) ~ 400M image-text pairs	Training Data: • PMC-15M ~ 15M biomed image-text pairs
Good for generalized tasks- suffers for fine-grained tasks like biomedical domain problems	Specialized on biomedical domain tasks.
Image Encoder : ResNet or ViT (Visual Image Transformer) (ViT-B/16. ViT-B/32 etc.)	Image Encoder: ViT-B/16
Text Encoder: Standard Transformer trained from scratch	Text Encoder: *PubMedBERT* , A BERT model trained on PubMed abstracts
Uses contrastive loss to align correct image-text pairs	Uses contrastive loss to align correct image-text pairs
Uses simple prompts like – *“An image of a glioma”*	Uses image-specific detailed prompts describing histology, anatomical structures etc
Struggles with specialized long vocabulary	Handles technical, multi-word compound terms. So a natural choice over CLIP when dealing with biomedical tasks with dynamic prompts

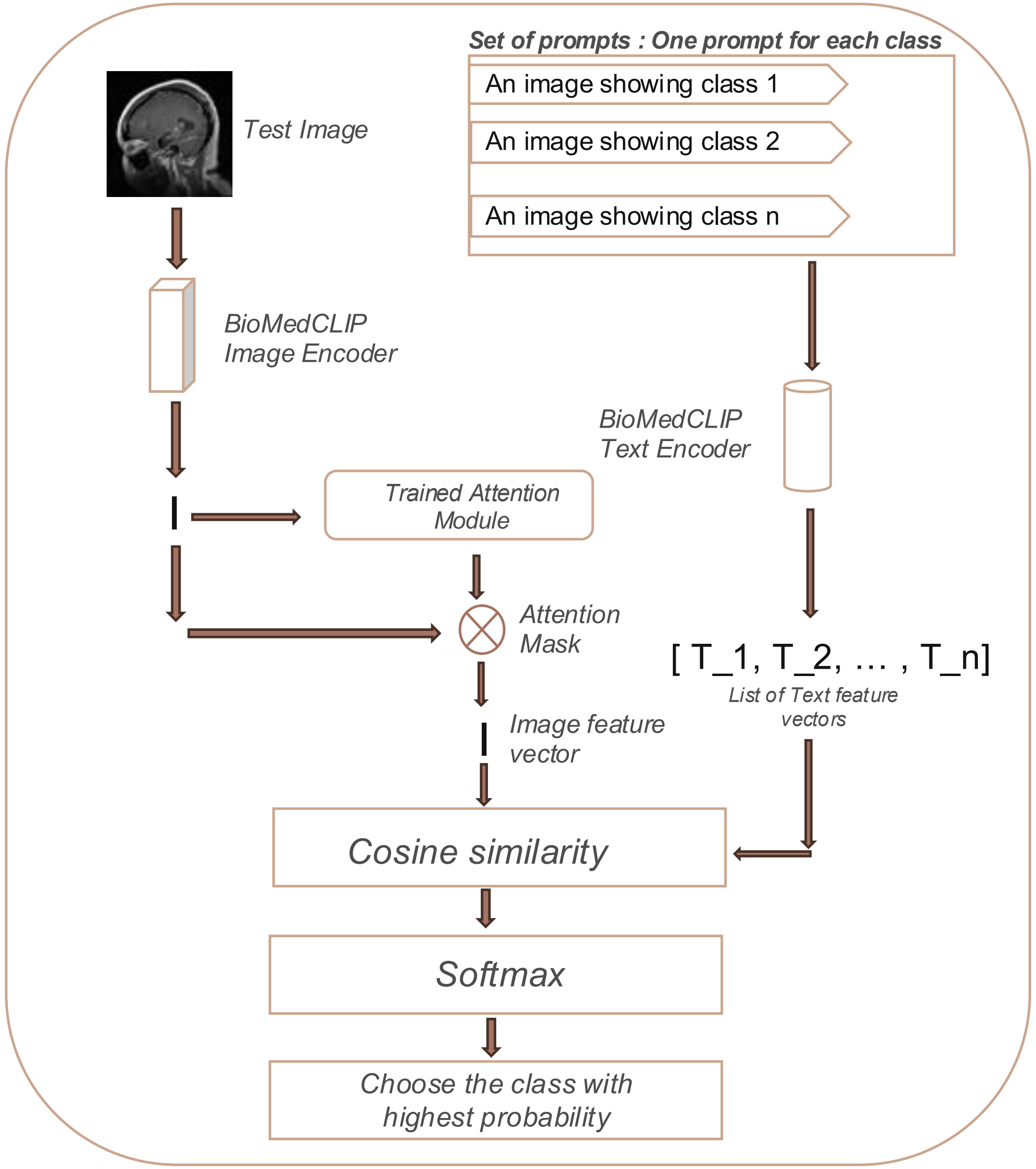
NOVEL IDEAS

Our training pipeline: Modified FACMIC architecture with BioMedCLIP encoders and dynamic prompt generation



NOVEL IDEAS

Inference Pipeline



NOVEL IDEAS

Experimental setup and hyperparameters

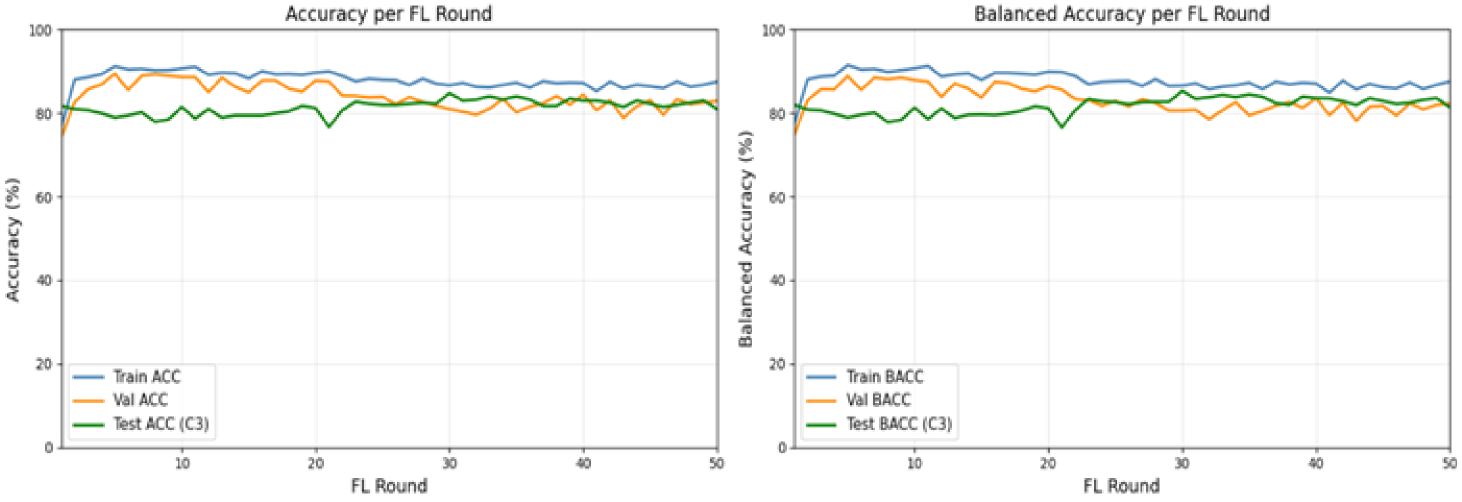
- **TRAINING CLIENTS**: Client 0, Client 1, Client 2
- **TEST CLIENTS/ Target domain**: Client 3
- **METRICS**: Accuracy, Balanced Accuracy
- **Backbone**: Frozen BioMedCLIP Encoders
- **ATTENTION MODULE**: 5 layers- Linear+ BN+ leakyReLU+linear+softmax
- **OPTIMIZER**: ADAM
- **BATCH SIZE**: 32
- **EPOCHS**: 50
- **LEARNING RATE**: 0.001
- **LAMBDA**: 1
- **GPU**: L4 GPU of Google Colab

NOVEL IDEAS

Results

```
==========================================================
 Final Results — best checkpoint on client_3 test set
==========================================================
 Test  ACC  = 84.77%
 Test  BACC = 85.32%

 Validation on training clients:
   client_0  ACC=86.30%  BACC=86.13%
   client_1  ACC=78.05%  BACC=78.33%
   client_2  ACC=81.91%  BACC=80.27%

 Best test ACC achieved at round 30
```

```
Round 28 | Global Attention Weight Norms:
  net.0.weight                          | norm = 1.615362
  net.0.bias                            | norm = 0.000411
  net.1.weight                          | norm = 3.321487
  net.1.bias                            | norm = 1.232780
  net.3.weight                          | norm = 3.332767
  net.3.bias                            | norm = 0.579224
Round 0028/50 | Train ACC=88.24%  Val ACC=82.70%  Test ACC=82.49%  Test BACC=82.70%
----------------------------------------------------------------
TRAINING FOR ROUND : 29
TRAINING FOR CLIENT : 0
TRAINING FOR CLIENT : 1
TRAINING FOR CLIENT : 2

Round 29 | Global Attention Weight Norms:
  net.0.weight                          | norm = 1.649221
  net.0.bias                            | norm = 0.000411
  net.1.weight                          | norm = 3.230951
  net.1.bias                            | norm = 1.241679
  net.3.weight                          | norm = 3.309429
  net.3.bias                            | norm = 0.587517
----------------------------------------------------------------
TRAINING FOR ROUND : 30
TRAINING FOR CLIENT : 0
TRAINING FOR CLIENT : 1
TRAINING FOR CLIENT : 2

Round 30 | Global Attention Weight Norms:
  net.0.weight                          | norm = 1.601108
  net.0.bias                            | norm = 0.000411
  net.1.weight                          | norm = 3.160023
  net.1.bias                            | norm = 1.262693
  net.3.weight                          | norm = 3.320818
  net.3.bias                            | norm = 0.595016
Round 0030/50 | Train ACC=86.67%  Val ACC=80.97%  Test ACC=84.77%  Test BACC=85.32%
  [*] Best model saved (ACC=84.77%)
```



Findings:

- BioMedCLIP with dynamic prompts has achieved the best accuracy of 84.77% and best balanced accuracy of 85.32%. It is better than the FACMIC results on this data
- The better performance may be due to better image-text alignment due to the detailed prompts and using BioMedCLIP instead of the CLIP model

CONCLUSION

In this work, we extended the FACMIC framework by addressing two key limitations of conventional vision-language alignment in biomedical federated settings.

1. I replaced **static, handcrafted prompts** with **dynamic prompt generation using LLaMA 3.2**. Unlike fixed prompts, which fail to capture variability across medical images and client distributions, dynamic prompts adapt to input-specific semantics, enabling richer and more context-aware textual representations.
2. I substituted the generic **CLIP** model with **BioMedCLIP**, which is pre-trained on domain-specific corpora such as PubMed and biomedical image-text pairs. This significantly improves cross-modal alignment in medical contexts where terminology, visual patterns, and semantics differ from natural images.

Together, these enhancements lead to:

-Improved **semantic alignment** between image and text embeddings

-Better **generalization across heterogeneous clients** in federated learning

-Increased robustness to **domain shifts and unseen medical variations**

FUTURE WORKS

1. **Evaluate on augmented dataset -** we plan to evaluate our setup on augmented datasets — incorporating translation, rotation, stain shifts, MRI variations, and synthetic data — to validate whether dynamic prompts better adapt to real-world clinical data heterogeneity compared to fixed prompt baselines.

1. **Partial Encoder Fine-Tuning with LoRA** Instead of keeping encoders fully frozen, apply **Low-Rank Adaptation (LoRA)** to the last few transformer blocks of the ViT encoder. LoRA adds very few trainable parameters, keeping communication costs low while allowing the encoder to adapt to medical imaging.

1. **Multi-Head Feature Attention** Replace the current single attention mask with a **multi-head attention mechanism** (similar to transformer attention), where each head focuses on different aspects of the image features — texture, shape, colour — before merging. This could capture richer task-specific features.

REFERENCES

1. Learning Transferable Visual Models From Natural Language Supervision- Radford et al - https://arxiv.org/abs/2103.00020

1. BiomedCLIP: a multimodal biomedical foundation model pretrained from fifteen million scientific image-text pairs - Sheng Zhang et al 2023 :https://arxiv.org/abs/2303.00915

1. The Llama 3 Herd of Models- Grattafiori et al : https://arxiv.org/abs/2407.21783

THANK YOU